The Coin-Toss Protocol

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Abstract

We present the Verifiable Coin Toss (VCT) protocol, a consensus mechanism for the WorldLand blockchain that combines the asynchronous, communication-free properties of Nakamoto consensus with energy consumption comparable to Proof-of-Stake (PoS) systems. In every block, each staked node independently and locally evaluates a Verifiable Random Function (VRF) over the previous block header to decide whether it earns the right to attempt Error Correction Code Proof-of-Work (ECCPoW). With a network-tunable acceptance probability p, only an expected pN of N active nodes engage in mining per block, reducing energy consumption by up to two orders of magnitude relative to Bitcoin while preserving the Nakamoto longest-chain rule. We derive an analytical expression for the adversarial block-win probability and prove that, under standard cryptographic assumptions together with ECCPoW's ASIC resistance and a hardware-bound device identifier (DID), a sustained majority attack requires the adversary to control more physical nodes than the honest network. We further demonstrate, both analytically and via Monte Carlo simulation matched to four decimal digits, that removing the DID requirement collapses VCT security exactly to stake-proportional PoS. To restore Bitcoin's "one CPU, one vote" property without specialized hardware deployment, we anchor node identity in TPM 2.0 endorsement keys, which are universally present on commodity systems. Finally, we introduce a tiered participation model incorporating NVIDIA H100-class GPU attestation, extending the protocol from a consensus mechanism into a marketplace for verifiable confidential AI computation. A reference implementation has been completed and deployed to a public beta network as of March 2026; the present paper formalizes its design and provides the underlying security analysis.

Index Terms

Blockchain, Nakamoto consensus, Verifiable Random Function, Proof-of-Work, ECCPoW, Sybil resistance, Trusted Platform Module, Confidential Computing, GPU attestation.

I. Introduction

The blockchain trilemma—the apparent impossibility of simultaneously achieving decentralization, security, and scalability—has driven the field through two dominant consensus paradigms. Proof-of-Work (PoW) systems exemplified by Bitcoin [1] provide unparalleled censorship resistance but consume energy at industrial scale. Proof-of-Stake (PoS) systems such as Ethereum's Casper [2] achieve dramatic energy reduction at the cost of replacing physical-world cost with capital-weighted voting, raising concerns about long-term decentralization and the loss of "one CPU, one vote" as articulated in the original Nakamoto whitepaper.

This paper introduces the Verifiable Coin Toss (VCT) consensus protocol, designed for the World-Land blockchain, which occupies a previously unexplored region of this design space. The central insight is that mining work need not be performed by every node simultaneously: if each block grants only a small subset of nodes the cryptographic right to attempt PoW, energy consumption scales with the size of this subset rather than the network size, while security remains anchored in the full network through the verifiable randomness of the selection process.

VCT formalizes this idea as follows. Each staked node, in every block, locally evaluates a VRF over the previous block hash, producing a pseudorandom output that is interpreted as a coin toss with bias p. Heads grants permission to compete in the ECCPoW [3], [4] race; tails forces the node to skip the block and wait. Crucially, this decision is made independently by each node without any communication, in keeping with the Satoshi Nakamoto [1] ideal of asynchronous emergent consensus. There is no committee, no Byzantine agreement protocol, no voting round. The system maintains the simplicity of Bitcoin while reducing the active mining population by the factor p.

A. Contributions

We make the following contributions:

1) We formalize the VCT protocol as a permission-gated extension of ECCPoW, preserving Nakamoto-style fork-choice rules while reducing per-block energy consumption by a factor of $1/p$ (typically 10^{-2} to 10^{-3}).
2) We prove that 51% attack resistance holds if and only if the adversary commits at least as many physical nodes as the honest network, given hardware-anchored DID and ECCPoW's ASIC resistance.
3) Through analytical derivation and Monte Carlo simulation matched to four decimal digits, we quantify the security degradation when hardware DID is replaced by standard PoS staking. The simplified system collapses exactly to stake-proportional PoS, with the additional vulnerability that hardware compute advantage compounds with stake.
4) We propose TPM 2.0 endorsement key attestation [7] as the practical mechanism for hardware-bound DID. Unlike prior proposals that depend on Intel SGX (now deprecated for consumer hardware) or specialized devices (Worldcoin's Orb), TPM 2.0 is universally deployed on modern commodity systems and required by Windows 11.
5) We extend the protocol with a tiered participation model that incorporates optional NVIDIA H100-class GPU attestation [8] via a hardware Root of Trust. This transforms idle node compute into a verifiable confidential computing marketplace, opening WorldLand to applications in healthcare AI, financial modeling, and government workloads that are currently dominated by centralized cloud providers.
6) We document the implementation status of the reference client, with the consensus engine deployed to a public beta network as of March 2026, and outline the remaining roadmap toward mainnet migration and confidential computation extension under the Rockies Upgrade series.

B. Paper Organization

Section II reviews the relevant primitives. Section III specifies the VCT protocol. Section IV presents the formal security analysis. Section V reports simulation results, including the consequences of removing hardware DID. Section VI describes TPM-anchored DID. Section VII introduces the GPU attestation tier. Section VIII outlines an implementation roadmap. Section IX discusses related work, and Section X concludes.

II. Background

A. Nakamoto Consensus and ECCPoW

In Nakamoto consensus [1], nodes append blocks to the chain by repeatedly hashing block headers until a target difficulty is met. The longest (or heaviest) chain is canonical. Security stems from the assumption that no adversary controls a majority of total hash power.

ECCPoW [3] replaces Bitcoin's SHA-256-based puzzle with a syndrome-decoding problem derived from low-density parity-check (LDPC) codes. Two properties are essential to the present work: (i) puzzles are time-varying, with parameters re-derived at each block; and (ii) memory-bandwidth requirements bound the speedup achievable by application-specific integrated circuits (ASICs), so per-machine hashrate ratios remain close to unity across commodity hardware. The recently proposed ECCVCC framework [4] extends this analysis with formal guarantees on ASIC resistance under a stronger threat model.

B. Verifiable Random Functions

A Verifiable Random Function [5] is a keyed pseudorandom function that, given a secret key sk and an input x, produces an output y together with a non-interactive proof π such that any party knowing the public key pk can verify $y = \mathrm{VRF}(sk, x)$. Crucially, y is indistinguishable from uniform random for a party who does not know sk, yet π proves to any verifier that y was correctly computed.

VRFs have been used for committee-based sortition in Algorand [6]. We use them differently: not to elect a leader or a committee, but to grant each node an independent right-to-mine decision.

C. Trusted Platform Module 2.0

TPM 2.0 [7] is a tamper-resistant cryptographic coprocessor specified by the Trusted Computing Group and standardized as ISO/IEC 11889. Each TPM contains a manufacturer-fused Endorsement Key (EK) whose public component is signed by the manufacturer's Root Certificate Authority. Cryptographic operations involving the EK execute exclusively within the TPM die; the private key is non-extractable except via invasive chip-level attack. TPM 2.0 is required by Windows 11 and present in essentially all PCs manufactured since 2021, including discrete chips from Infineon, Nuvoton, and STMicroelectronics, as well as firmware-based implementations in Intel and AMD CPUs.

D. GPU Confidential Computing

NVIDIA's Hopper architecture (H100, 2022) [8] introduced an on-die hardware Root of Trust supporting a Trusted Execution Environment (TEE) with measured boot and remote attestation. Each H100 contains a fuse-burned Identity Key whose certificate is signed by NVIDIA's Root CA. The architecture continues in H200 and Blackwell-generation GPUs (B100/B200). This is conceptually a TPM equivalent specialized for accelerator devices, although it is presently confined to data-center-class GPUs rather than consumer cards.

III. The VCT Protocol

A. System Model

Let $\mathcal{N}$ denote the set of nodes participating in WorldLand, with $|\mathcal{N}| = N$. Each node $i \in \mathcal{N}$ maintains a long-term VRF key pair (sk_i, pk_i) and a hardware-anchored device identifier DID_i bound to pk_i

(Section VI). To participate, node i posts a stake s_0 (the protocol parameter, e.g., 10^5 WL tokens) into a stake-management contract on the WorldLand chain, binding the stake to DID_i.

We denote by H_h the canonical block header at height h. The protocol maintains a network-tunable parameter $p \in (0,1)$, reset epoch-wise to keep the expected number of mining-eligible nodes per block at a target value T, typically $T \in [10^2, 10^3]$.

B. Verifiable Coin Toss Function

The core primitive is the Verifiable Coin Toss function:

$$\mathrm{VCT}_i(H_{h-1}) = 1\left[\, \mathrm{VRF}(sk_i, H_{h-1} \,\|\, h) < p \cdot 2^{|y|} \,\right], \tag{1}$$

where y is the VRF output, $|y|$ is its bit-length, and $1[\cdot]$ is the indicator function. Heads ($\mathrm{VCT}_i = 1$) entitles node i to attempt ECCPoW for block h; tails ($\mathrm{VCT}_i = 0$) requires the node to skip the block.

The construction has three essential properties:

1) **Locality.** Equation (1) is computable by node i alone, requiring only sk_i and the public H_{h-1}. No interaction with other nodes is needed.
2) **Unbiasedness.** The pseudorandomness of VRF ensures that, conditional on sk_i, the output is statistically independent of H_{h-1}. Conversely, the dependence on H_{h-1} prevents grinding attacks that would otherwise let a miner search over alternative previous blocks.
3) **Verifiability.** The accompanying VRF proof π_i allows any other node, given pk_i and H_{h-1}, to verify that the coin was tossed honestly without revealing sk_i.

C. Block Production and Validation

The mining loop at node i proceeds as follows:

Algorithm 1 VCT-Gated Block Production at Node i

Wait until H_{h-1} is finalized
$(y_i, \pi_i) \leftarrow \mathrm{VRF}(sk_i, H_{h-1} \| h)$
if $y_i / 2^{|y|} < p$ **then** ▹ heads
  Begin ECCPoW search for block h
  if nonce ν found such that $\mathrm{ECCPoW}(\nu, H_{h-1}, T_h) = \text{valid}$ **then**
    Construct block H_h with fields $(\nu, pk_i, y_i, \pi_i, \mathrm{DID}_i)$
    Broadcast H_h
  end if
else ▹ tails
  Skip block h, wait for H_h from peers
end if

A receiving node validates a candidate H_h by checking, in order: (i) pk_i is registered to a staked DID; (ii) $\mathrm{Verify}(pk_i, H_{h-1}\|h, y_i, \pi_i) = 1$; (iii) $y_i/2^{|y|} < p$; and (iv) the ECCPoW solution ν is valid against H_{h-1} at the prevailing difficulty T_h. Fork choice follows the heaviest-ECCPoW-work rule of standard Nakamoto consensus, with VRF output as a deterministic tiebreaker.

D. Dynamic p Adjustment

At each epoch boundary (e.g., every 10 000 blocks), p is recomputed as

$$p \;\leftarrow\; \frac{T}{N_{\text{active}}}, \tag{2}$$

where N_{active} is the number of accounts that have successfully mined or submitted a valid heads proof in the preceding window. This estimator avoids the difficulty-adjustment ambiguity of pure PoW chains: the network maintains an exact on-chain count rather than a statistical estimate.

IV. Security Analysis

A. Adversary Model

Let N_H and N_A denote the number of nodes controlled by the honest network and the adversary, respectively. Each node holds at most one DID, and each DID is bound to exactly one stake of s_0 tokens. The adversary may collude internally but is otherwise restricted by the cryptographic assumptions below.

Assumption 1 (VRF Security)**.** *The VRF satisfies existential unforgeability under chosen-message attacks (EUF-CMA) and computational pseudorandomness.*

Assumption 2 (ECCPoW ASIC Resistance)**.** *For commodity hardware, the per-machine ECCPoW solution rate h_i satisfies $h_A/h_H \in [1, c]$ for a small constant c (typically $c \leq 3$).*

Assumption 3 (Hardware-Bound DID)**.** *Each DID is cryptographically anchored to a unique physical hardware root of trust, such that producing K distinct valid DIDs requires K distinct hardware components.*

B. Block Win Probability

We derive the probability that the adversary mines the next block. Let $X_H \sim \text{Binomial}(N_H, p)$ and $X_A \sim \text{Binomial}(N_A, p)$ denote the number of honest and adversarial mining-eligible nodes in a given block. Conditional on $(X_H, X_A) = (x_H, x_A)$ with $x_H + x_A > 0$, the probability that the adversary wins the ensuing ECCPoW race is

$$\Pr[A \text{ wins} \mid x_H, x_A] = \frac{h_A x_A}{h_A x_A + h_H x_H}. \tag{3}$$

Taking expectations and noting that for large N_H, N_A the binomial distributions concentrate around their means,

$$\Pr[A \text{ wins}] \approx \frac{h_A N_A}{h_A N_A + h_H N_H}. \tag{4}$$

Critically, the parameter p cancels: VCT does not introduce stake- or probability-weighting beyond what is implied by node count.

C. Sustained Majority Attack

Theorem 1 (51% Attack Resistance)**.** *Under Assumptions 1–3, an adversary can sustainedly produce more than half of the chain's blocks if and only if*

$$N_A > \frac{h_H}{h_A} N_H. \tag{5}$$

In particular, when $h_A = h_H$ (the asymptotic case under strong ASIC resistance), the condition reduces to $N_A > N_H$.

Proof. Equation (4) expresses the per-block win probability. By the Strong Law of Large Numbers, the long-run fraction of blocks attributable to the adversary converges almost surely to this ratio. Solving $\Pr[A \text{ wins}] > 1/2$ for N_A yields (5). The reverse direction follows by symmetry. □

Proposition 1 (Linear Attack Cost). *Let C_{node} denote the marginal cost of one valid (DID, stake) tuple. The minimum cost of a sustained majority attack is*

$$C_{attack} \geq \frac{h_H}{h_A} N_H \cdot C_{node}. \tag{6}$$

That is, attack cost scales linearly with the size of the honest network, in contrast to capital-weighted PoS where stake concentration via secondary markets can compress the cost.

V. Consequences of Removing Hardware DID

A natural design simplification is to replace hardware-bound DID with standard PoS staking only: each account holds stake s_0 and obtains one VCT toss per block, but no physical-hardware constraint binds accounts to machines. This section quantifies the security impact.

A. *Modified Adversary Model*

Without hardware DID, an adversary can host multiple staked accounts on a single physical machine. Let M_A denote the number of adversarial machines, each hosting K accounts, so that the total adversarial account count is $N_A = M_A K$ and the total adversarial stake is $S_A = M_A K s_0$.

A machine has multiple opportunities to be eligible per block but can only run one ECCPoW search at a time, since the puzzle is memory-bound. The probability that machine j is eligible (at least one of its K accounts gets heads) is $q = 1 - (1 - p)^K$, and conditional on eligibility, its contribution to the ECCPoW race is one machine-equivalent.

B. *Block Win Probability*

Generalizing (4):

$$\Pr[A \text{ wins}] \approx \frac{h_A M_A \big(1 - (1 - p)^K\big)}{h_A M_A \big(1 - (1 - p)^K\big) + h_H N_H p}. \tag{7}$$

C. *Optimal Adversary Strategy*

For fixed total stake budget S_A, the constraint $M_A K = S_A / s_0$ is invariant. Differentiating (7) with respect to K at fixed $M_A K$ shows that the maximum is attained at $K = 1$, that is, one account per machine. Stacking accounts on fewer machines yields strictly worse attack effectiveness because the per-machine eligibility probability saturates at $q \to 1$, but each additional account beyond the first on a given machine is wasted.

D. *Equivalence to Pure PoS at $K = 1$*

Substituting $K = 1$ into (7) and assuming $h_A = h_H$:

$$\Pr[A \text{ wins}] = \frac{M_A p}{M_A p + N_H p} = \frac{S_A}{S_A + S_H}. \tag{8}$$

Theorem 2 (PoS Equivalence Without DID). *Without hardware-bound DID, and under the optimal adversary strategy $K = 1$, the VCT protocol's block-win probability is identical to pure stake-proportional PoS.*

TABLE I

MONTE CARLO VERIFICATION OF ANALYTICAL BLOCK-WIN PROBABILITIES AT ADVERSARY STAKE = HONEST STAKE.

K	Analytical	Monte Carlo	|Diff|
1	0.5000	0.4999	1×10^{-4}
10	0.4944	0.4948	4×10^{-4}
100	0.4409	0.4416	7×10^{-4}
1000	0.1657	0.1681	2.4×10^{-3}

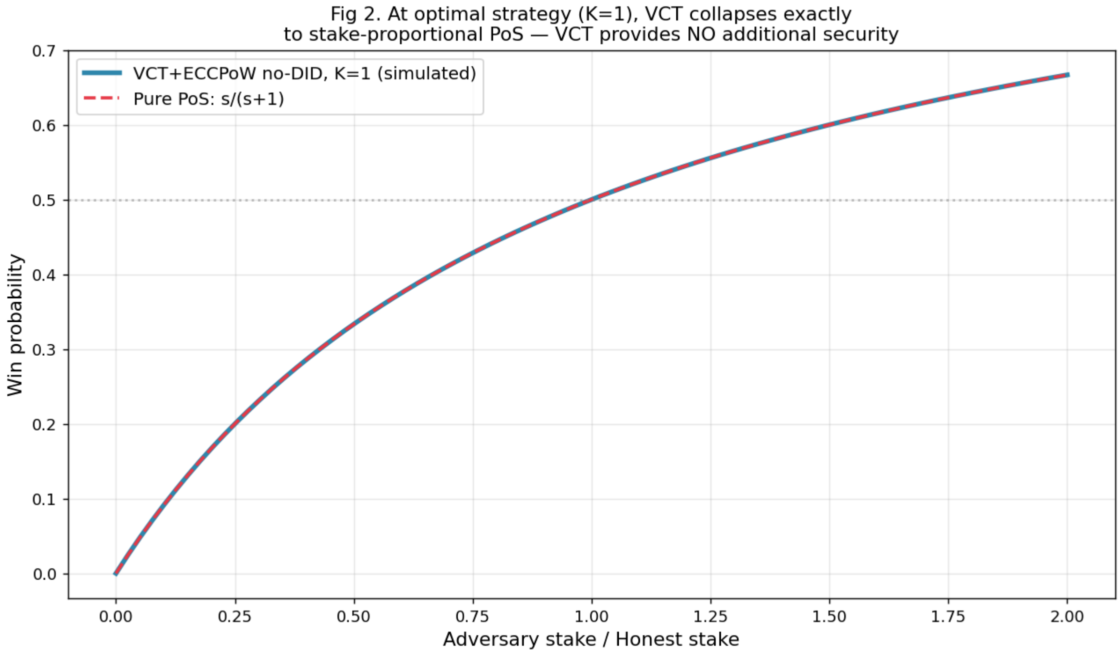

Fig. 1. Without hardware-bound DID, the optimal adversary strategy ($K = 1$) makes VCT win-probability identical to pure stake-proportional PoS. The simulated curve for VCT+ECCPoW (blue) coincides with the analytical PoS curve $s/(s+1)$ (red dashed) to within numerical precision.

E. Simulation Verification

We verified (7) and Theorem 2 by Monte Carlo simulation over $N_H = 10\,000$ honest machines, target $T = 100$, and adversary stake fractions in $\{0.25, 0.5, 1.0, 1.5\}$, with 50 000 blocks per configuration. Table I reports the agreement.

F. Hardware Advantage as Residual Risk

Equation (7) also reveals that adversary hardware advantage h_A/h_H multiplies stake fraction directly. The stake required for a 51% attack is reduced to $1/(h_A/h_H)$ of the honest stake (Fig. 2). For example, $h_A/h_H = 2$ halves the required stake. ECCPoW's ASIC resistance is the principal defense against this vector; if the resistance is broken in any future analysis, no-DID VCT becomes substantially weaker than pure PoS.

VI. TPM-ANCHORED DEVICE IDENTITY

The simulations of Section V establish that hardware-bound DID is a load-bearing assumption. We now describe a concrete construction using TPM 2.0 that is deployable on commodity hardware

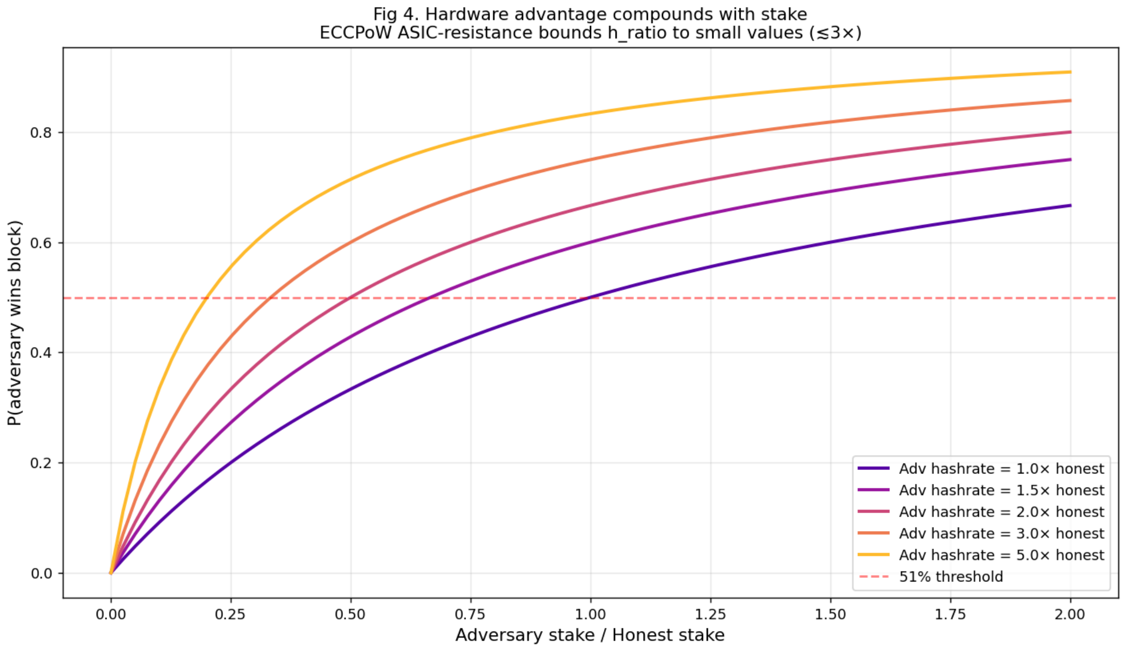

Fig. 2. Hardware compute advantage compounds with stake fraction in the no-DID regime. Each additional unit of h_A/h_H proportionally reduces the stake required for a 51% attack. ECCPoW's ASIC resistance bounds h_A/h_H near unity, but this remains the dominant residual attack vector.

without specialized provisioning.

A. *Why TPM 2.0*

Several alternatives were considered and rejected:

- *Intel SGX* (used by Hyperledger Sawtooth's PoET): deprecated for consumer CPUs since 2021; numerous side-channel vulnerabilities discovered between 2018 and 2022.
- *Worldcoin Orb*: proof of unique personhood via biometric hardware; requires global rollout of dedicated devices.
- *Software fingerprinting* (CPU ID, MAC address): trivially spoofable.

By contrast, TPM 2.0 chips are present in essentially all modern PCs as a Windows 11 requirement and are installed as either discrete chips or firmware in Intel/AMD CPUs. Multiple manufacturers (Infineon, Nuvoton, STMicroelectronics) sustain a competitive supply, avoiding single-vendor dependency.

B. *DID Registration Protocol*

The protocol binds a node's VRF public key to its TPM Endorsement Key chain.

C. *Privacy via zk-Attestation*

Direct on-chain disclosure of Cert_{EK} reveals the TPM manufacturer and serial number, raising privacy concerns. We propose wrapping the registration in a zk-SNARK that proves possession of a valid certificate chain without disclosing the certificate itself. The SNARK predicate is

$$\exists\,(\text{Cert}_{\text{EK}}, \sigma)\ :\ \text{ChainVerify}(\text{Cert}_{\text{EK}}, \mathcal{R})\ \wedge\ \text{Verify}_{\text{AIK}}(\sigma, pk)$$

Algorithm 2 TPM-Anchored DID Registration

1: Node generates VRF key pair (sk, pk) inside TPM with non-extractable attribute
2: TPM derives an Attestation Identity Key (AIK) from the EK
3: Node obtains Cert_{EK} from TPM manufacturer
4: TPM produces $\sigma \leftarrow \text{Sign}_{\text{AIK}}(pk \,\|\, \text{nonce})$
5: Node submits transaction (DID = $H(pk)$, pk, Cert_{EK}, σ, stake)
6: Smart contract verifies:
- Cert_{EK} chains to a known TPM manufacturer Root CA
- σ is a valid AIK signature derived from Cert_{EK}
- $H(\text{Cert}_{\text{EK}}.\text{serial})$ has not been previously registered (anti-reuse)

7: On success, contract records DID $\leftrightarrow$ pk $\leftrightarrow$ stake binding

where $\mathcal{R}$ is the set of accepted manufacturer Root CA public keys, hard-coded into the circuit. This construction integrates with the recursive ZKP infrastructure already developed for WorldLand's Data Availability Layer.

D. Tiered Participation

To accommodate hardware diversity, three tiers are defined:

TABLE II
Tiered participation model

Tier	Hardware	Stake	VCT Weight
Full	TPM 2.0 EK attestation	s_0	1×
Enhanced	TPM + SGX/SEV-SNP	s_0	1×, higher finality
Light	No hardware attestation	$10s_0$	1×

The Light tier compensates for the absence of hardware-bound Sybil resistance through increased stake requirements, recovering an economic equivalent of physical scarcity. As the network matures, the relative weight of Full-tier rewards can be increased to encourage migration toward stronger Sybil resistance.

VII. GPU Attestation Tier for Verifiable AI Compute

The protocol described thus far secures consensus. We now extend it to monetize the substantial GPU compute resources that participating nodes possess but typically leave idle outside of mining. This extension transforms WorldLand from a single-purpose blockchain into a marketplace for verifiable confidential AI computation, addressing a market currently dominated by centralized cloud providers.

A. GPU as a Trusted Execution Environment

NVIDIA H100 and subsequent generations integrate an on-die hardware Root of Trust capable of generating signed attestation reports proving that (i) the GPU is a genuine NVIDIA product, (ii) the loaded firmware (VBIOS, GSP firmware) matches an approved measurement in the Reference Integrity Manifest, and (iii) confidential computing mode is enabled. Verification is performed via the NVIDIA Remote Attestation Service (NRAS) or, for self-sovereign deployments, by directly checking signatures against the published NVIDIA Root CA.

B. Composite Attestation

For end-to-end confidentiality, GPU attestation must be combined with CPU-side TEE attestation (Intel TDX or AMD SEV-SNP). The composite quote covers:

- CPU TEE measurements (kernel, drivers, application code)
- GPU TEE state (firmware, configuration mode)
- Encrypted communication channel between CPU TEE and GPU TEE

The Intel Trust Authority and equivalent services provide unified attestation tokens that can be referenced on-chain.

C. Marketplace Design

A confidential workload provider posts a job to the WorldLand chain specifying (i) the desired attestation requirements, (ii) the maximum compute time, (iii) payment escrowed in WL tokens. Eligible nodes (those with attested H100-class hardware) bid for the job. The selected provider executes the workload within the joint CPU/GPU TEE, and posts the encrypted output along with a fresh attestation quote tying the output to the same hardware. Settlement releases the escrowed payment if the attestation verifies.

This construction has the property that the workload provider need not trust any single party: the verification chain anchors in NVIDIA and Intel/AMD Root CAs, both well-established commercial entities, and the on-chain record provides non-repudiable evidence.

D. Limitations

We note three limitations of GPU attestation:

- *Vendor concentration*. NVIDIA's Root CA represents a single point of trust for the GPU TEE. We mitigate this by additionally accepting AMD MI300-class attestation and remaining open to future vendors.
- *Consumer hardware exclusion*. Consumer GPUs (e.g., RTX 4060/5090) lack confidential computing. The tier system accommodates this by relegating such nodes to Basic and GPU-Miner tiers without confidential compute participation.
- *Attestation freshness*. Attestation quotes have validity windows; long-running jobs may require periodic re-attestation, with associated overhead.

VIII. Implementation Status and Roadmap

A. Reference Implementation Status

The protocol described in this paper has been instantiated as a reference implementation within the WorldLand client, an EVM-compatible fork of go-ethereum maintained at https://github.com/cryptoecc/WorldLand. The implementation has progressed through two stages prior to the present writing in May 2026:

Stage 1 (January 2026): Initial integration of the Verifiable Coin Toss function into a development testnet, with block headers extended to carry the VRF output and proof. Per-epoch VRF evaluation and batched verification were implemented to amortize the cryptographic overhead.

Stage 2 (March 2026): Deployment of a consolidated VRF-gated ECCPoW consensus engine—comprising VCT evaluation, the LDPC-based puzzle continuing the ECCPoW lineage of [3], [4], difficulty adjustment, sealing, and JSON-RPC interfaces—to a public beta network operated separately from the existing Seoul mainnet.

All implementation work is presently undergoing senior review by the INFONET Laboratory research team prior to acceptance into the master branch of the reference client. The remaining roadmap, organized into three concurrent tracks under the *Rockies Upgrade* series, is summarized below.

B. Track A: Mainnet Migration — Rockies Upgrade v.1.0 (May – September 2026)

A1. Senior code review (May – June 2026). Comprehensive review and refinement of the consensus engine implementation by senior researchers, with particular attention to fork-choice correctness, VRF integration soundness, and adherence to the protocol specification of Section III.

A2. Beta network stress test (May – August 2026). Expansion of public beta participation to validate liveness, fork rate, and energy efficiency claims at scale.

A3. External security audit (July – August 2026). Independent audit by a recognized blockchain security firm, conducted partially in parallel with A2.

A4. Coordinated mainnet hardfork (September 2026). Activation of the VCT protocol on the Seoul mainnet at a predetermined block height, constituting the first official release of the **Rockies Upgrade v.1.0** milestone and the inaugural increment of the new mountain-range hardfork series.

C. Track B: Hardware-Anchored Identity Layer — Rockies Upgrade v.1.1 (July 2026 – February 2027)

B1. TPM attestation client (July – August 2026). Integration of TPM 2.0 endorsement key verification into the node software using established open-source libraries.

B2. DID registry (August – October 2026). On-chain registry binding device identifiers to VRF public keys and stake, with anti-reuse logic via Endorsement Certificate serial number commitments.

B3. Privacy-preserving attestation (October 2026 – January 2027). A zk-SNARK construction proving valid TPM ownership without disclosing the manufacturer certificate, building upon the recursive ZKP infrastructure developed for the WorldLand Data Availability Layer.

B4. Tier activation (February 2027). Governance-approved differential reward structure for Full, Enhanced, and Light participation tiers, activated as the **Rockies Upgrade v.1.1** milestone.

D. Track C: Confidential Computation Marketplace — Rockies Upgrade v.1.2 (October 2026 – December 2027)

C1. Resource scheduler (October – December 2026). Cooperative scheduling between consensus participation and tenant workloads via Kubernetes integration, allowing nodes to monetize compute capacity outside of the active mining phase.

C2. Workload marketplace (January – March 2027). Off-chain order book with on-chain settlement; escrow contracts denominated in WL tokens; bidding and matching protocol for confidential workloads.

C3. GPU attestation integration (March – June 2027). Support for NVIDIA H100/H200/B100 hardware attestation, AMD MI300 SEV-SNP attestation, and composite CPU–GPU attestation tokens issued through Intel Trust Authority or self-hosted equivalents. Activation of GPU-attested confidential compute as the **Rockies Upgrade v.1.2** milestone.

C4. Pilot deployments (June – December 2027). Confidential AI workload pilots with healthcare, financial, and government tenants, transitioning the marketplace from technical readiness to production operation.

E. Schedule Summary

TABLE III
TRACK SCHEDULE AND ROCKIES UPGRADE MILESTONES (MAY 2026 BASELINE).

Track	Start	On-chain activation	Completion
A: Mainnet migration	May 2026	Sep 2026 (*v.1.0*)	Sep 2026
B: Identity layer	Jul 2026	Feb 2027 (*v.1.1*)	Q1 2027
C: Compute marketplace	Oct 2026	Jun 2027 (*v.1.2*)	Q4 2027

The Rockies Upgrade series formally inaugurates a new hardfork era for the WorldLand mainnet. Versions v.1.0 through v.1.2 collectively transition the network from pure ECCPoW operation through VCT-gated mining, hardware-anchored Sybil resistance, and ultimately a marketplace for verifiable confidential computation.

IX. RELATED WORK

Algorand [6] pioneered VRF-based sortition for committee selection; we adopt the VRF primitive but apply it to per-node mining permission rather than committee membership, retaining Nakamoto's communication-free architecture. Hyperledger Sawtooth's Proof of Elapsed Time [9] used Intel SGX to enforce randomized waiting times but was largely confined to permissioned deployments and depends on now-deprecated SGX support.

Hardware-attestation-based Sybil resistance has been proposed in academic settings (Truxen [10], zk-PoI [11]) but has not seen production deployment in any major permissionless chain. Worldcoin [12] pursues an orthogonal approach using biometric proof-of-personhood, requiring deployment of specialized hardware.

The ECCPoW puzzle family [3], [4], on which VCT relies for its competitive mining phase, has been analyzed extensively in prior work, with formal results on ASIC resistance arising from the time-varying nature of the LDPC syndrome decoding problem.

Our distinguishing contribution is the simultaneous combination of (i) Nakamoto-style asynchronous mining, (ii) energy efficiency comparable to PoS, (iii) Sybil resistance via universally deployed commodity hardware (TPM 2.0), and (iv) a tiered extension to verifiable confidential AI compute. To our knowledge, no prior or contemporary protocol unifies these four properties.

X. CONCLUSION

We have presented the Verifiable Coin Toss (VCT) protocol, a consensus mechanism that augments ECCPoW with VRF-based mining permission to achieve Bitcoin-like security at PoS-level energy consumption. We proved that, under hardware-bound DID, sustained 51% attack requires an adversarial node count exceeding the honest network. Conversely, removing hardware DID collapses the protocol exactly to stake-proportional PoS, recovering capital-weighted security but losing the "one CPU, one vote" property of Bitcoin. We proposed TPM 2.0 endorsement key attestation as the practical mechanism for hardware-bound DID, leveraging hardware that is universally deployed on modern commodity systems. Finally, we extended the protocol with a tiered model incorporating NVIDIA H100-class GPU attestation, opening WorldLand to verifiable confidential AI computation as a marketplace orthogonal to consensus.

Future work will address: (i) formal verification of the protocol in TLA+ or Coq; (ii) detailed economic modeling of the marketplace tier under competing cloud pricing; (iii) full implementation of the post-quantum VRF, including a hybrid scheme during the transition period; and (iv) empirical measurement of h_A/h_H across modern commodity hardware, which determines the residual security margin in the no-DID regime.

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